Waves at boundaries

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe reflection and transmission at a boundary.

SUCCESS CRITERIA - I can...

1. Identify incident, reflected and transmitted waves.
2. Distinguish fixed-end from free-end pulse reflection.
3. Track energy without assuming equal amplitudes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “boundary” and “reflection” to explain your answer. Sentence starters are optional.

boundary: the place where a medium or constraint changes

reflection: a wave returning from a boundary

transmission: a wave continuing through or beyond a boundary

01 Warm-up: bring back an earlier idea

A reflected signal returns after **3 s**. **Name the extra quantity needed** to **calculate distance** and **explain why this** is a round trip.

Word help: signal = a disturbance that carries information

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

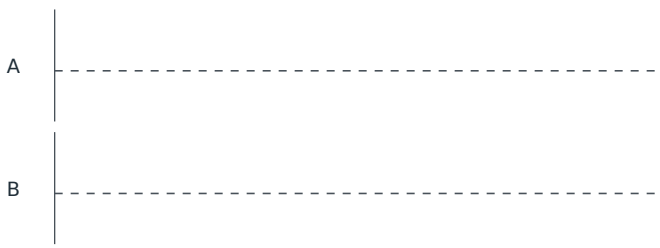
02 Worked example: explain the prediction

Reference: $f_{reflected} = f_{incident}$.
At a stationary boundary. A displacement pulse returns inverted from an ideal fixed end and upright from an ideal free end.

Worked reasoning: At a fixed end, an upward rope pulse reflects downward. At an ideal free end it reflects without inversion. The end of the rope is held differently in these two cases.

Your turn: Draw incident and reflected pulse snapshots for both ends; **show propagation directions**.

Sentence starter: In my work, I used … to show … . I labeled it … .



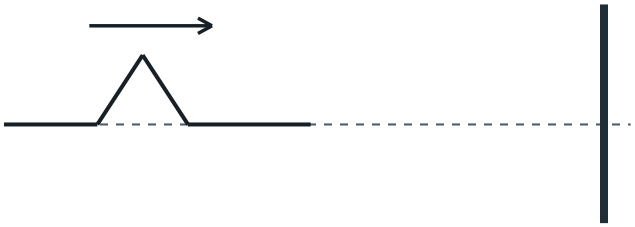
Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

An upward rope pulse reaches an ideal fixed end. **Sketch the reflected pulse** and its travel direction.

Word help: ideal = using stated simplifying conditions

Sentence starter: In my work, I used … to show … . I labeled it … .



Sketch and label your response.

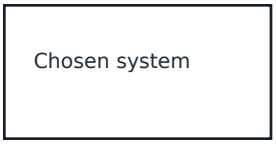
Marzano check - circle: ① ② ③

04 Apply the physics

A pulse carries **10 J** toward a boundary. The reflected pulse carries **4 J**, and no energy is absorbed. **Complete the energy-flow diagram** and **find the transmitted energy**.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Does a smaller transmitted amplitude prove frequency changed? **Explain what usually fixes the** frequency at a stationary boundary.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** sample completely changes phase with latent heat **600 J/kg**. **Find the energy magnitude required**. **Label temperatures**, samples and relevant energy transfers.

Word help: latent heat = energy per mass needed for a specified phase change

Sentence starter: I found ... [unit]. This value means ... in this situation.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

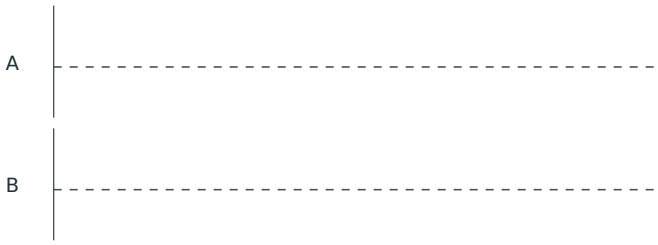
Marzano check - circle: 1 2 3

07 Exit check

For an ideal free-end rope reflection, is an upward displacement pulse returned upright or inverted?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice ... . This supports ... because ... .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two equal upward pulses approach each other on an ideal rope.
1. **Sketch overlap** and later separation.
2. **Explain whether** they permanently stick.

Word help: ideal = using stated simplifying conditions

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Sketch and label your response.

Marzano check - circle: 1 2 3